

Set Theory and Posets

Sets, Operations, Inclusion–Exclusion, and Hasse Diagrams

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Outline

- 1 Introduction to Sets
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What is a Set?

Definition

A **set** is a well-defined collection of distinct objects, referred to as elements or members of the set.

Key Concepts:

- If x is an element of set A , we write $x \in A$.
- If x is not an element of set A , we write $x \notin A$.
- Sets are typically denoted by capital letters (A, B, C), while elements are denoted by lowercase letters (a, b, c).

Representing Sets: Set-Builder Form

Sets can be described in two primary ways:

- 1 **Roster/Tabular Form:** Explicitly listing elements.
 - Example: $A = \{2, 4, 6, 8\}$
- 2 **Set-Builder Form:** Stating the property that all elements must satisfy.

General Notation

$$S = \{x \mid P(x)\} \quad \text{or} \quad S = \{x : P(x)\}$$

Read as: “The set of all x such that $P(x)$ is true.”

Examples:

- $E = \{x \in \mathbb{Z} \mid x \text{ is even and } 0 < x < 10\}$
- $B = \{x \in \mathbb{R} \mid x^2 - 4 = 0\}$ (which equals $\{-2, 2\}$)

Common Types of Sets

- **Empty (Null) Set ($\emptyset$ or $\{\}$):** A set containing no elements.
- **Singleton Set:** A set containing exactly one element, e.g., $\{5\}$.
- **Finite Set:** A set with a countable number of elements, e.g., $A = \{a, b, c\}$.
- **Infinite Set:** A set with infinitely many elements, e.g., $\mathbb{N} = \{1, 2, 3, \dots\}$.
- **Universal Set (U):** The set containing all possible objects under consideration.
- **Subset ($A \subseteq B$):** Every element of A is also an element of B .
- **Power Set ($\mathcal{P}(A)$):** The set of all subsets of A .
 - If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

Fundamental Set Operations

Union ($A \cup B$)

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Intersection ($A \cap B$)

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

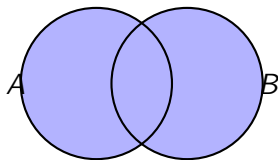
Difference ($A \setminus B$ or $A - B$)

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

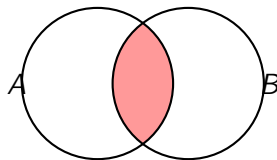
Complement (A^c or A')

$$A^c = U \setminus A = \{x \in U \mid x \notin A\}$$

Venn Diagrams: Union & Intersection



$$A \cup B$$



$$A \cap B$$

Inclusion–Exclusion Principle (PIE)

The Inclusion–Exclusion Principle helps compute the cardinality of the union of multiple sets by accounting for overlapping elements.

For Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For Three Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

PIE Example

Problem: In a class of 50 students, 30 play Football, 25 play Basketball, and 10 play both. How many students play at least one game?

Solution: Let F be the set of Football players and B be the set of Basketball players.

- $|F| = 30$
- $|B| = 25$
- $|F \cap B| = 10$

$$|F \cup B| = |F| + |B| - |F \cap B|$$

$$|F \cup B| = 30 + 25 - 10 = 45$$

Answer: 45 students play at least one game.

Posets and Hasse Diagrams

Definition

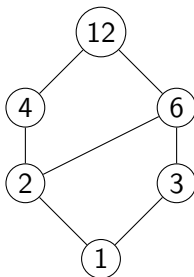
A **Partially Ordered Set (Poset)** is a pair $(S, \leq)$ where S is a set and $\leq$ is a binary relation that is *reflexive*, *antisymmetric*, and *transitive*.

What is a Hasse Diagram?

- A simplified visual representation of a finite poset.
- **Rules for construction:**
 - 1 Omit self-loops (implied by reflexivity).
 - 2 Omit transitive edges (e.g., if $a \leq b$ and $b \leq c$, do not draw $a \rightarrow c$).
 - 3 Draw edges without directional arrows; place larger elements higher up.

Hasse Diagram Example

Example: Hasse Diagram for the divisibility relation on $D_{12} = \{1, 2, 3, 4, 6, 12\}$.



Thank You!